

Problem

Solve the following differential equations subject to the specified initial conditions

(a) $d^2v/dt^2 + 4v = 12$, $v(0) = 0$, $dv(0)/dt = 2$

(b) $d^2i/dt^2 + 5 di/dt + 4i = 8$, $i(0) = -1$, $di(0)/dt = 0$

(c) $d^2v/dt^2 + 2 dv/dt + v = 3$, $v(0) = 5$, $dv(0)/dt = 1$

(d) $d^2i/dt^2 + 2 di/dt + 5i = 10$, $i(0) = 4$, $di(0)/dt = -2$

Step-by-step solution

Step 1 of 12

(a)

$$\frac{d^2V}{dt^2} + 4V = 12, \quad V(0) = 0$$

$$\text{and } \frac{dV(0)}{dt} = 2$$

$$\frac{1}{RC} = \text{Coefficient of } \frac{dv}{dt} = 0$$

$$\Rightarrow \alpha = \frac{1}{2RC} = 0$$

$$\frac{1}{LC} = 4$$

Step 2 of 12

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = 2$$

$$\frac{V_s}{LC} = 12$$

$$\Rightarrow V_s = \frac{12}{4}$$

$$V_s = 3$$

$$\therefore \alpha < \omega_0 \text{ [Under damped]}$$

Step 3 of 12

So,

$$V(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$[\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 2]$$

$$V(t) = 3 + (A_1 \cos 2t + A_2 \sin 2t) \quad [e^{-0} = 1]$$

At $t = 0$;

$$V(0) = 3 + A_1 = 0$$

$$\Rightarrow A_1 = -3$$

$$\frac{dV(t)}{dt} = -2 A_1 \sin 2t + 2 A_2 \cos 2t$$

$$\Rightarrow \frac{dV(0)}{dt} = 2 = 2 A_2$$

$$\Rightarrow A_2 = 1$$

Or, $V(t) = (3 - 3 \cos 2t + \sin 2t) \text{ V}$

Step 4 of 12

(b)

$$\frac{d^2 i}{dt^2} + 5 \frac{di}{dt} + 4i = 8$$

$$i(0) = -1;$$

$$\frac{di(0)}{dt} = 0$$

$$\frac{1}{RC} = 5$$

$$\Rightarrow \alpha = \frac{1}{2RC}$$

$$\alpha = \frac{5}{2}$$

$$\alpha = 2.5$$

Step 5 of 12

$$\frac{1}{LC} = 4$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \sqrt{4}$$

$$\omega_0 = 2$$

$$\frac{i_s}{LC} = 8$$

$$\Rightarrow i_s = \frac{8}{4}$$

$$i_s = 2$$

Step 6 of 12

$\therefore \alpha > \omega_0$ [Over damped]

So, $i(t) = i_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_1 = -2.5 + \sqrt{(2.5)^2 - 2^2}$$

$$s_1 = -1$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -2.5 - \sqrt{(2.5)^2 - 2^2}$$

$$s_2 = -4$$

Step 7 of 12

So, $i(t) = 2 + A_1 e^{-t} + A_2 e^{-4t}$

At $t = 0$;

$$i(0) = -1$$

$$i(0) = 2 + A_1 + A_2$$

$$\Rightarrow A_1 + A_2 = -3 \rightarrow (1)$$

Step 8 of 12

$$\frac{dV(t)}{dt} = -A_1 e^{-t} - 4A_2 e^{-4t}$$

$$\Rightarrow \frac{dV(0)}{dt} = 0 = -A_1 - 4A_2$$

$$\Rightarrow A_1 + 4A_2 = 0 \rightarrow (2)$$

On solving equations (1) and (2) we get,

$$A_2 = 1 \quad \text{And} \quad A_1 = -4$$

So, $i(t) = (2 - 4e^{-t} + e^{-4t}) A$

Step 9 of 12

(c)

$$\frac{d^2V}{dt^2} + 2\frac{dV}{dt} + V = 3,$$

$$V(0) = 5;$$

$$\frac{dV(0)}{dt} = 1$$

$$\frac{1}{RC} = 2 \Rightarrow$$

$$\alpha \Rightarrow \frac{1}{2RC} = \frac{2}{2} = 1$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_0 = 1 = \left[\because \frac{1}{LC} = 1 \right]$$

$$\frac{V_s}{LC} = 3$$

$$\Rightarrow V_s = 3$$

$$\therefore \alpha = \omega_0 = [\text{Critically damped}]$$

Step 10 of 12

So,

$$V(t) = V_s t (A_1 + A_2 t) e^{-\alpha t}$$

$$V(t) = 3 + (A_1 + A_2 t) e^{-t}$$

$$\frac{dV(t)}{dt} = -A_1 e^{-t} + A_2 (e^{-t} - e^{-t} t)$$

$$\Rightarrow \frac{dV(0)}{dt} = 1$$

$$\frac{dV(0)}{dt} = -A_1 + A_2$$

$$\Rightarrow A_2 = 2 + 1 = 3$$

$$\text{Hence, } V(t) = 3 = (2 + 3t) e^{-t} \text{ V}$$

Step 11 of 12

(d)

$$\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 5i = 10$$

$$i(0) = 4$$

$$\frac{di(0)}{dt} = -2$$

$$\frac{1}{RC} = 2$$

$$\Rightarrow \alpha = \frac{1}{2RC} = 1$$

$$\frac{1}{LC} = 5$$

Step 12 of 12

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \sqrt{5}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\omega_d = \sqrt{5-1}$$

$$\omega_d = 2$$

$$\frac{i_s}{LC} = 10 \Rightarrow i_s = \frac{10}{5} = 2$$

$\therefore \alpha < \omega_0$ [Under damped],

So,

$$i(t) = i_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$i(t) = 2 + (A_1 \cos 2t + A_2 \sin 2t) e^{-t}$$

$$i(0) = 4 = 2 + A_1 \Rightarrow A_1 = 2$$

$$\frac{di(t)}{dt} = (-2 A_1 \sin 2t + 2 A_2 \cos 2t) e^{-t} + (A_1 \cos 2t + A_2 \sin 2t)(-e^{-t})$$

$$\Rightarrow \frac{di(0)}{dt} = -2 = 2 A_2 - A_1 \Rightarrow A_2 = 0$$

Hence, $i(t) = 2 + 2 \cos 2t e^{-t}$ A